Liang (Twist) Shan

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Research Interest

- Statistical foundations, optimization, and mechanistic interpretability of **Modern Machine Learning**, with a focus on deep learning, reinforcement learning, and generative models including language models and diffusion models.

Education

- **Peking University** — **Beijing, China**
  B.S. in Statistics, School of Mathematical Sciences — *Jul. 2024–Jul. 2028 (Expected)*
  Second Major B.A. in Economics, National School of Development — *Sep. 2025–Jul. 2028 (Expected)*
- **University of Copenhagen** — **Copenhagen, Denmark**
  International Exchange Student, Department of SCIENCE — *Aug. 2025–Jan. 2026*
- **Peking University** — **Beijing, China**
  Undergraduate Studies in Physics, School of Earth and Space Sciences — *Sep. 2023–Jun. 2024*

Publications and Preprints

- **Beyond Scalar Leaderboards: Adaptive Sampling for Pareto Frontier Identification in Multi-Objective LLM Arenas** [arXiv] [Blog] [Code]
  Liang Twist Shan, *Jiahao Ai, Zhimei Ren* — *Under review at ICLR 2027*
- **Targeted Retrieval, Compact Representations: How CoT Reasoning Improves Long-Context Counting** [arXiv] [Blog] [Code]
  Liang Twist Shan, *Tianyu Hu, Hao Yan, Yiqiao Zhong* — *Under review at ICLR 2027*

Research Experience

- **Statistical Modeling for Prompt-Dependent LLM Routing** — **University of Pennsylvania**
  Advisor: Prof. Zhimei Ren, UPenn — *Nov. 2026 – Present*
  - Develop statistical models for prompt-dependent LLM routing from pairwise preferences, exploring prompt and model representations and trade-offs among response quality, cost, and latency.
- **Safety of LLM Agents and Multi-Agent Systems** — **University of Wisconsin–Madison**
  Advisor: Prof. Yiqiao Zhong, UW–Madison — *Oct. 2026 – Present*
  - Investigate how LLM agents maintain safety-critical information in internal states and external memory, and whether monitoring these states can detect hidden failures or overlooked requirements.
  - Study whether implicit communication among LLM agents can combine distributed information to produce potentially unsafe collective outcomes, even without explicit coordination or joint optimization.
- **Muon for Diffusion: Theory and Denoising Dynamics** — **University of Pennsylvania**
  Advisor: Prof. Yuxin Chen, UPenn — *Sep. 2026 – Present*
  - Study the convergence and spectral learning dynamics of Simplified Muon in diffusion denoising, combining theoretical analysis of linear models with controlled experiments on nonlinear denoising networks.
- **Mechanistic Interpretability of Latent Reasoning in DLMs** — **University of California, Berkeley**
  Advisor: Prof. Song Mei, UC Berkeley — *Aug. 2026 – Present*
  - Investigate latent reasoning in diffusion language models through controlled tasks, causal interpretability, and fine-tuning, examining how model architecture and decoding order shape internal computation.
- **Pareto Frontier Identification from Pairwise Preferences** — **University of Pennsylvania**
  Advisor: Prof. Zhimei Ren, UPenn — *May. 2026 – Oct. 2026*
  - Introduce an order-preserving vector-Borda reduction and adaptive elimination algorithm for fixed-confidence Pareto frontier identification, with instance-dependent upper and minimax lower bounds on sample complexity.
  - Evaluate VB-EGE on synthetic benchmarks and Arena, SearchArena, and VisionArena data, reducing comparison cost by up to 18.3× relative to uniform BT-MLE.

- **Mechanistic Interpretability of Chain-of-Thought Reasoning** — **University of Wisconsin–Madison**
  Advisor: Prof. Yiqiao Zhong, UW–Madison — *Apr. 2026 – Sep. 2026*
  - Identify a mechanistic shift from broad retrieval in Non-thinking models to iterative targeted retrieval under chain-of-thought, accompanied by substantially more separable and compact internal count representations.
  - Establish causal pathway of two different mechanisms using attention-head ablation and activation patching, showing that evolving counter states guide subsequent retrieval and final count prediction.

Selected Honors and Awards

- **Yu Hao Dream Scholarship** — **Peking University**
  Scholarship supporting international exchange program — *May. 2026*

Skills & Others

- **Computer Skills**: Python (&PyTorch), R, C++ (Elementary), LaTeX
- **Languages**: Mandarin Chinese (Native), English (Fluent)
- **GRE General Test**: 325/340 (Verbal 155, Quantitative 170)
- **Interests**: Photography, Travel, Cooking, Basketball, Kayaking